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Circuits 2 lab 02
KUID: 2026962128

Experiment 2: Differential amplifiers

Objective: To understand the uses and physics behind differential amplifiers which are designed to amplify the difference between two signals. We seek to understand differential mode gain vs common mode to differential conversion. Lastly, we also will study feedback and use it in the experiment.

Materials: 1kOhm resistor x3, 1.2kOhm resistor x1, NMOS 2N7000 x 4, CS-147 transformer x 1

3.1 Differential signals

$$Vov = Vgs - Vth = (5 - 3.2675) - 1.6 = 0.1325V$$

$$gm1 = \frac{2ID}{Vov} = \frac{2\ x\ 8.12\ x\ 10^{-3}}{0.1325} = 122.57\ x\ 10^{-3}\ A/V$$

$$gm2 = \frac{2ID}{Vov} = \frac{2\ x\ 6.47\ x\ 10^{-3}}{0.1325} = 97.66\ x\ 10^{-3}\ A/V$$

$$gm = \frac{gm1 + gm2}{2} = 110.12\ x\ 10^{-3}\ A/V$$

| Rout [kΩ] | Ad [V/V] |
|---|---|
| 1 | 110.12 |

$A_d$ - $gmR_{out}$ = $110.12\ x\ 10^{-3}\ A/V$ x 1000 = 110.12 V

Voltage at which the distortion exists: 0-2V and 13-14V

3.3

| $V_{out+}$ [$V_{pp}$] | $V_{out-}$[$V_{pp}$] | $V_{out}$[$V_{pp}$] | $A_d$ [V/V] |
|---|---|---|---|
| 1.24 | 1.44 | 2.68 | 12.88 |

| $V_{CM}$ [V] | $V_{out}$[V] | $A_{CM-DM}$ [V/V] |
|---|---|---|
| 5V | 0.138 | 0.06 |
| 5.1 | 0.144 | |

$$ACM - DM = \frac{\Delta Vout}{\Delta VCM} = \frac{0.144-0.138}{5.1-5} = \frac{0.006}{0.1} = 0.06\,V/V$$

| CMRR [dB] |
|---|
| 46.64 |

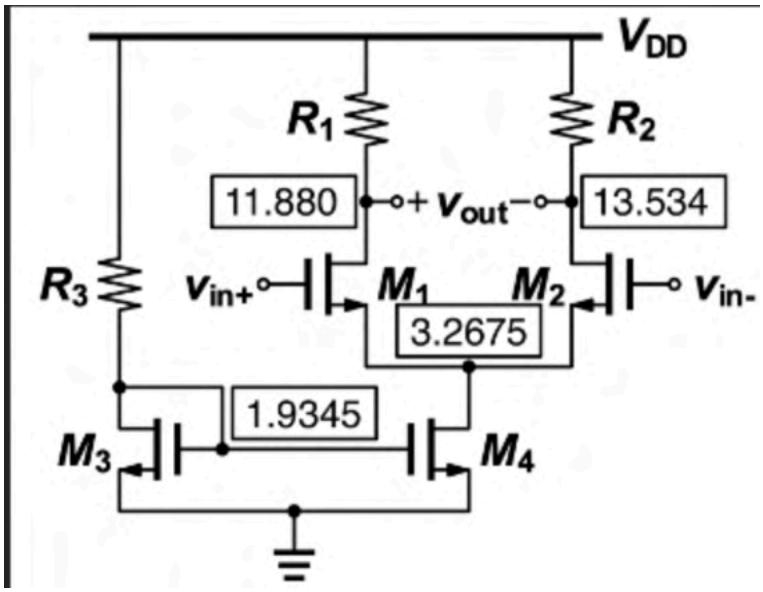
$$20log_{10}\left|\frac{Ad}{ACM-DM}\right| = 20log_{10}\left|\frac{12.88}{0.06}\right| = 46.64$$

3.1

| $V_{in}$0 [Vpp] |
|---|
| 0.208 |

| $V_{CM}$ + ½ $V_{in}$ [$V_{pp}$] | $V_{CM}$- ½ $V_{in}$ [$V_{pp}$] |
|---|---|
| 0.092 | 0.094 |

| $V_{in}$ [Vpp] | $V_{out+}$ [$V_{pp}$] | $V_{out-}$[$V_{pp}$] | $V_{out}$[$V_{pp}$] | $A_d$ [V/V] |
|---|---|---|---|---|
| 0.208 | 3.50 | 4.16 | 7.66 | 36.83 |



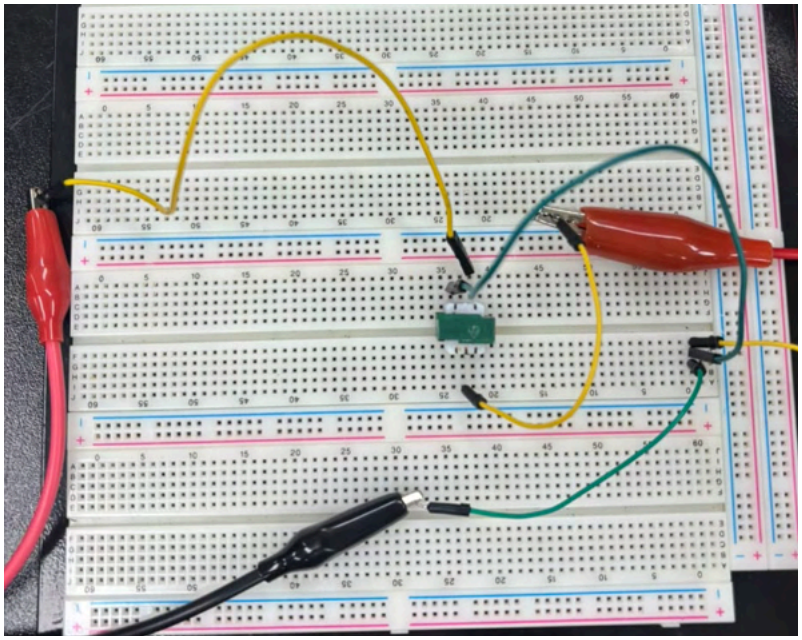

| $I_{D1}$ [mA] | $I_{D2}$ [mA] |
|---|---|
| 8.12 | 6.47 |

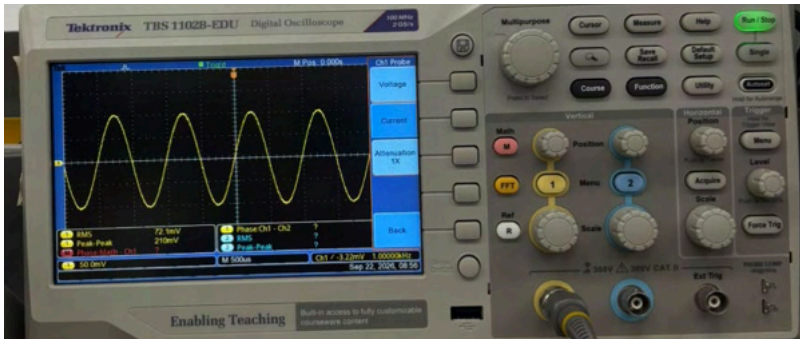
$$Id1 = \frac{VDD - 11.880}{R1} = 8.12\,mA$$

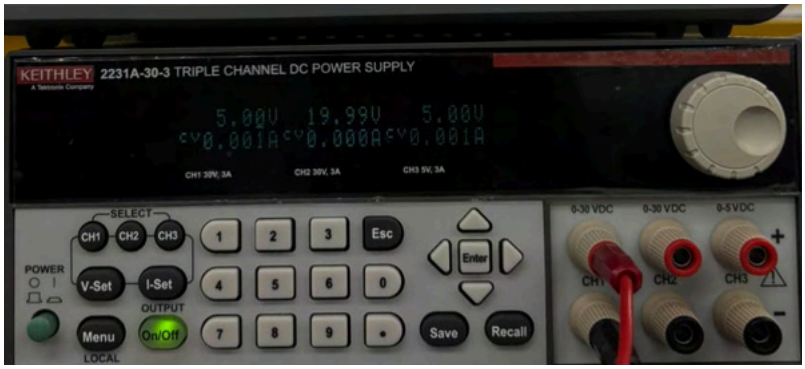
$$Id2 = \frac{VDD - 13.534}{R2} = 6.47\,mA$$

| $G_{m1}$ [mA/V] | $G_{m2}$ [mA/V] | $G_m$ [mA/V] |
|---|---|---|
| 112.57 | 97.66 | 110.12 |

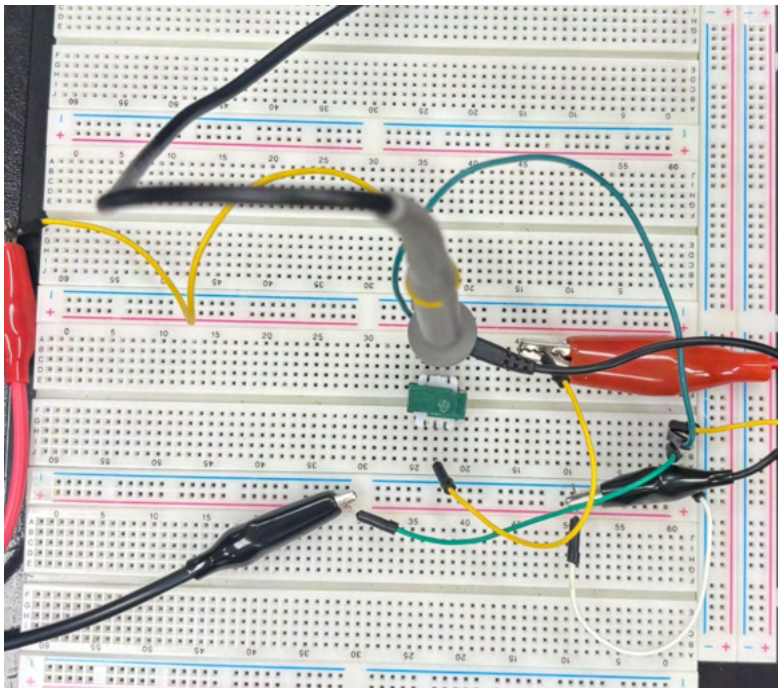
Photos:



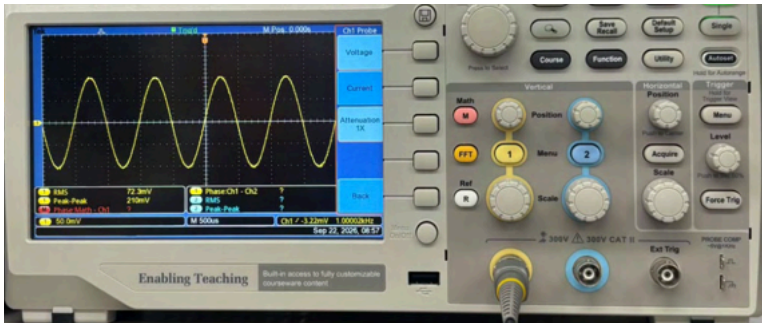

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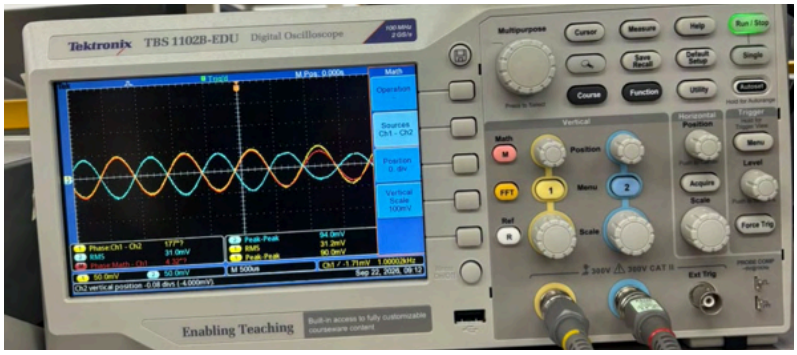

KEITHLEY
2231A-30-3 TRIPLE CHANNEL DC POWER SUPPLY
SELECT
CH1
CH2
CH3
V-Set
I-Set
OUTPUT
On/Off
Menu
LOCAL
POWER
Esc
Enter
Save
Recall
0-30 VDC
0-30 VDC
0-5 VDC

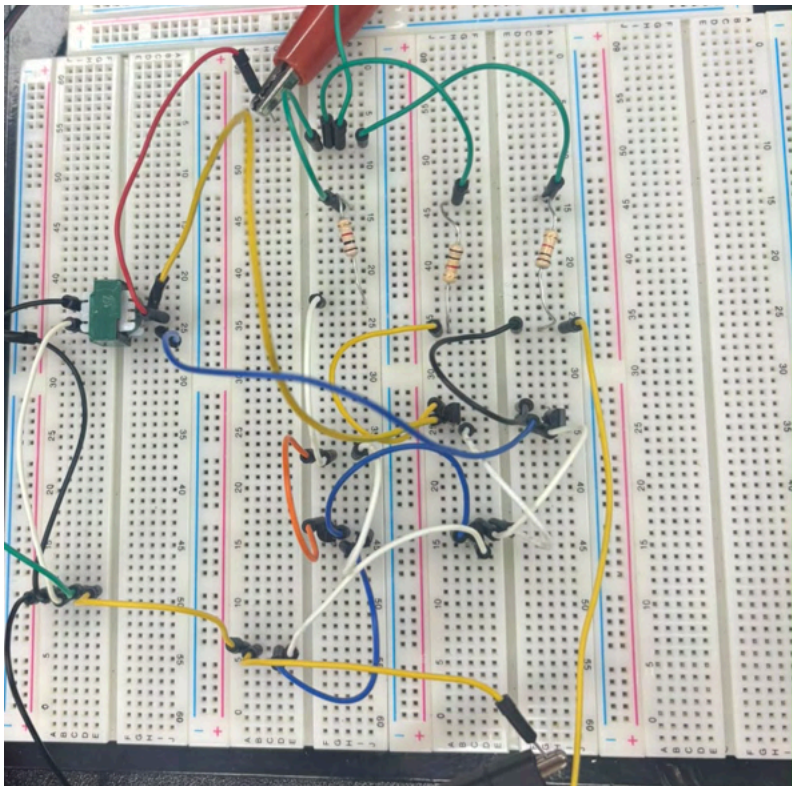


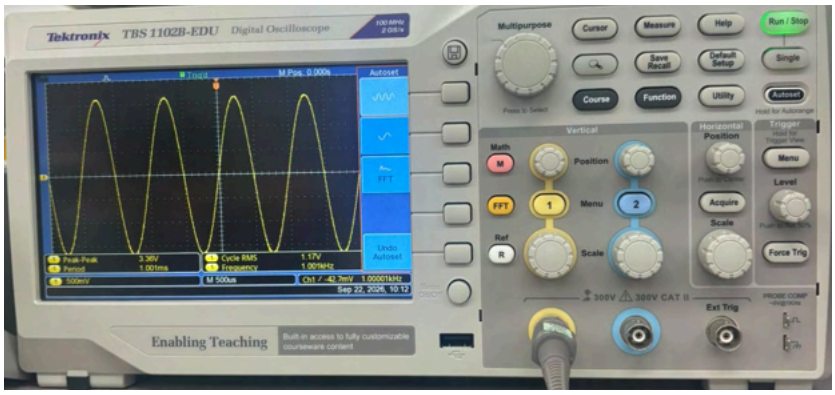



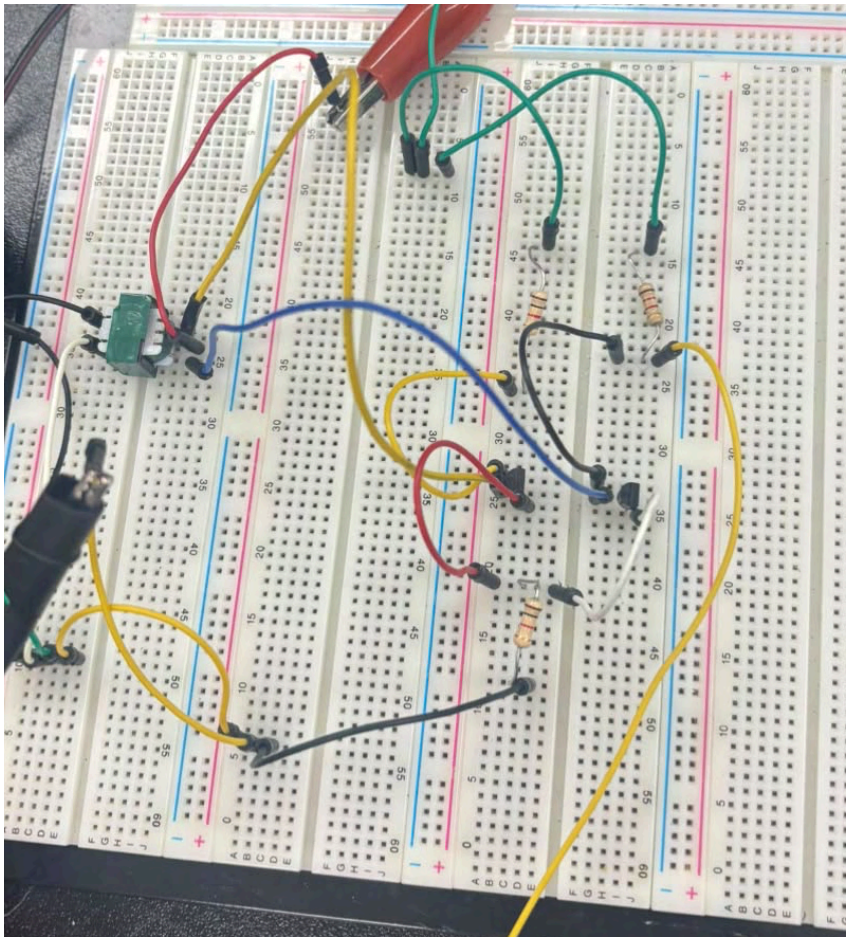
Next Experiment:

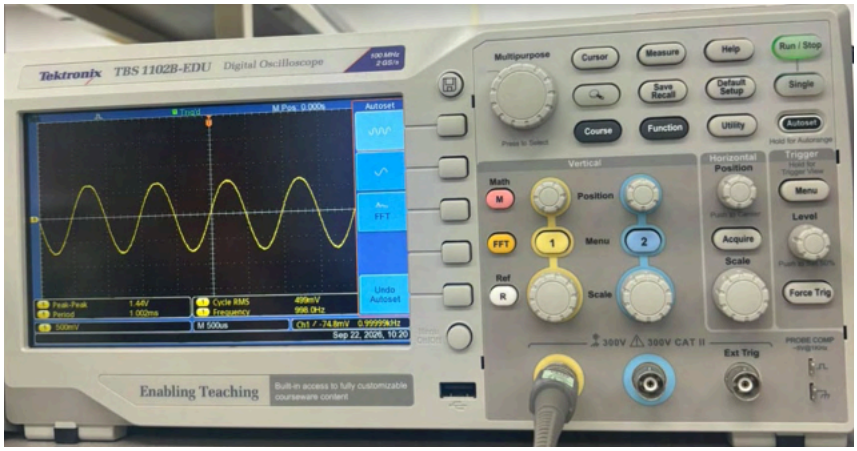


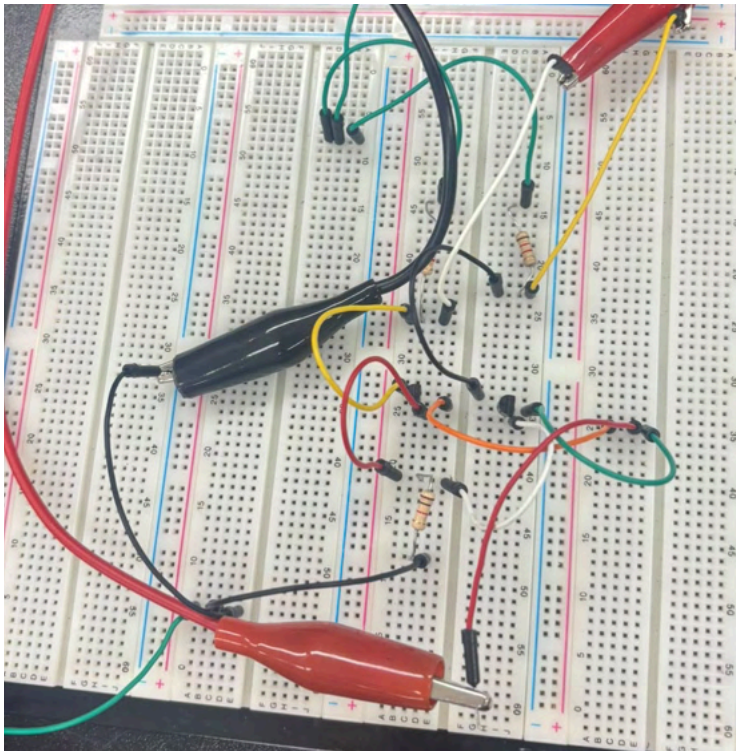

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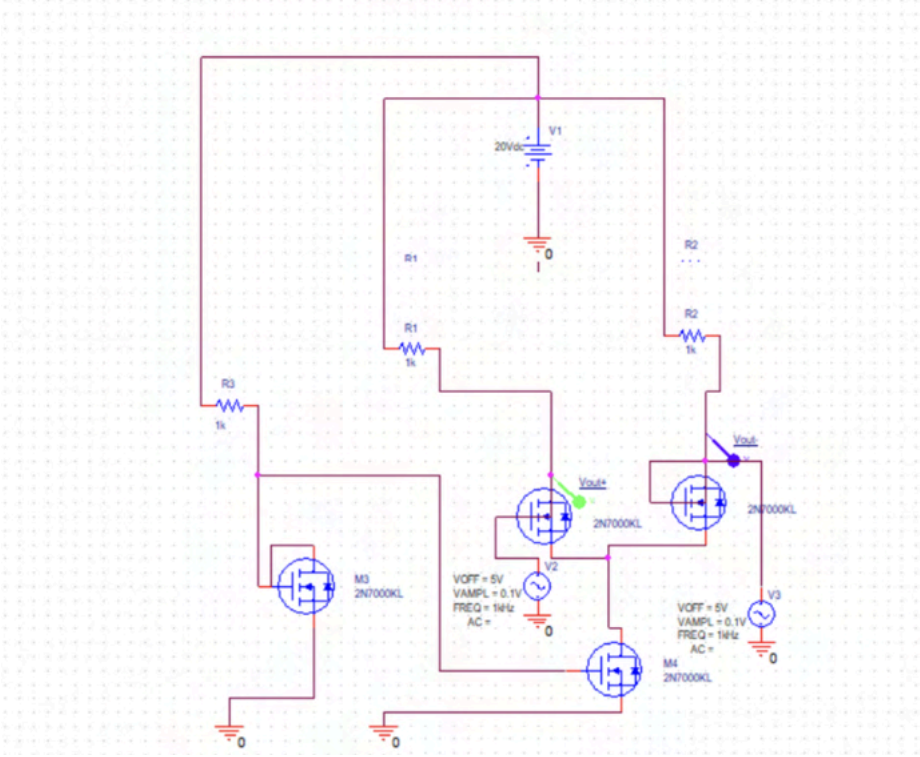

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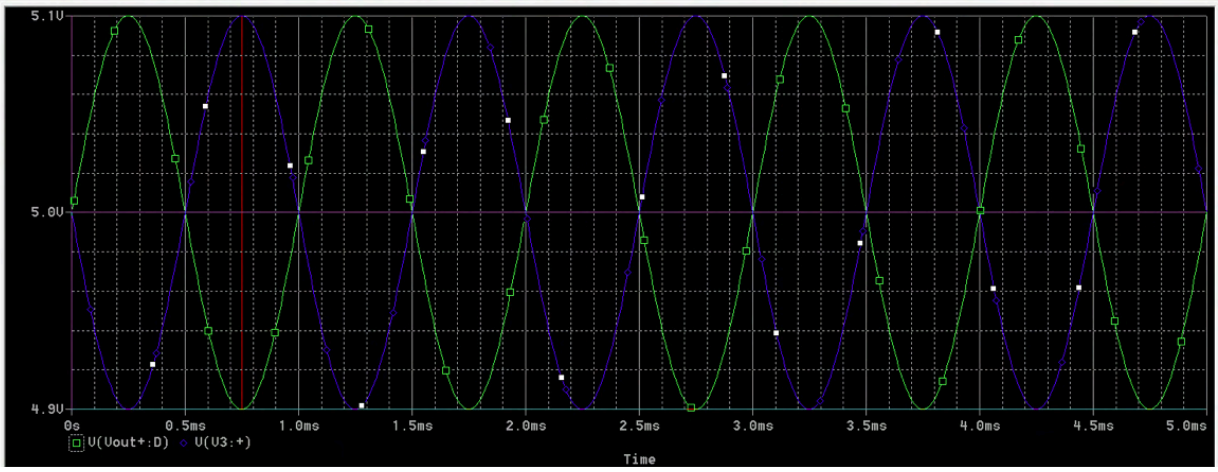



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Function
Utility
Autoset
Vertical
Horizontal Position
Trigger
Math
Position
Menu
Level
FFT
Acquire
Scale
Force Trig
Ref
Ext Trig
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PSPICE


V1
20Vdc
0
R1
R2
R1
1k
R2
1k
R3
1k
Vout-
Vout+
2N7000KL
2N7000KL
M3
2N7000KL
V2
VOFF = 5V
VAMPL = 0.1V
FREQ = 1kHz
AC =
0
V3
VOFF = 5V
VAMPL = 0.1V
FREQ = 1kHz
AC =
0
M4
2N7000KL
0
0


5.1V
5.0V
4.9V
0s
0.5ms
1.0ms
1.5ms
2.0ms
2.5ms
3.0ms
3.5ms
4.0ms
4.5ms
5.0ms
V(Vout+:D)
V(V3:+)
Time